

CHAPTER 2

FEASIBILITY STUDY

2.1 Technical Feasibility

The system is technically feasible using stable and widely supported tools. The selected technologies ensure performance, scalability, and maintainability.

Technology Stack:

- Backend: Node.js with Express
- Frontend: React.js
- Database: MongoDB
- AI Integration: Gemini API (for transcription, insight extraction, pattern analysis, PRD generation, and chat editing)

Key Points:

- Modular backend with pipeline design ensures easy debugging and maintenance.
- AI tasks are offloaded to Gemini, ensuring backend performance is not affected.
- Scalable architecture allows future upgrades or migration to managed cloud services.

2.2 Economic Feasibility

The economic feasibility evaluates the cost of development and operation compared to the expected benefits of the system.

Costs:

- Developer time for system development and maintenance
- Backend and frontend hosting
- Managed MongoDB hosting (optional)
- Gemini API usage charges

Benefits:

- Significant time savings in processing interviews and generating PRDs
- Faster decision-making and improved product quality
- Cost-effective due to use of open-source technologies and usage-based AI pricing

2.3 Operational Feasibility

The system integrates smoothly with the current workflow of product managers. The process follows a structured pipeline: interview upload → transcription → insight extraction → PRD generation → export or ticketing.

- Intuitive user interface with inline editing and chat assistance
- Minimal training required; only a basic introduction is needed
- Data stored securely in MongoDB for long-term retrieval and editing

2.4 Legal and Ethical Feasibility

The system ensures responsible handling of user data and compliance with applicable privacy standards.

- Data privacy ensured through authentication, database protection, and encryption
- User consent required before uploading recordings to AI services
- Compliance with external AI provider privacy policies is mandatory

2.5 Schedule Feasibility

The project timeline is realistic and manageable. The core system already demonstrates end-to-end functionality.

- Modular design enables addition of features without rewriting the codebase
- Development and extension schedule is predictable and well-structured